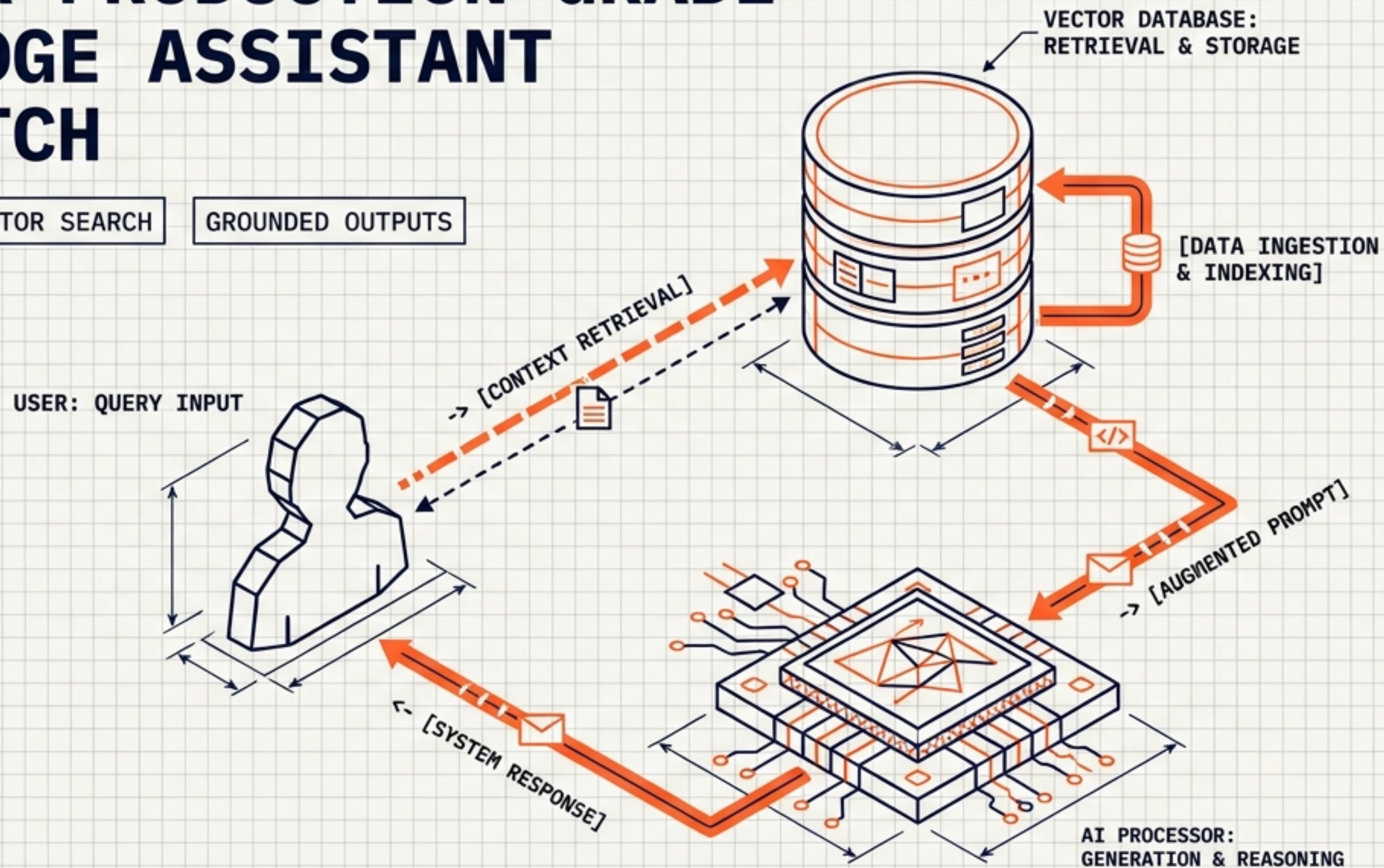


# BUILDING A PRODUCTION-GRADE AI KNOWLEDGE ASSISTANT FROM SCRATCH

ZERO CLOUD COST

LOCAL VECTOR SEARCH

GROUNDING OUTPUTS



# THE OLD WAY



# THE NEW WAY

Can I transfer to another campus if my average is below 80%?

Yes, but you must meet...

[SOURCE: Campus\_Transfer\_Policy\_v2.pdf]

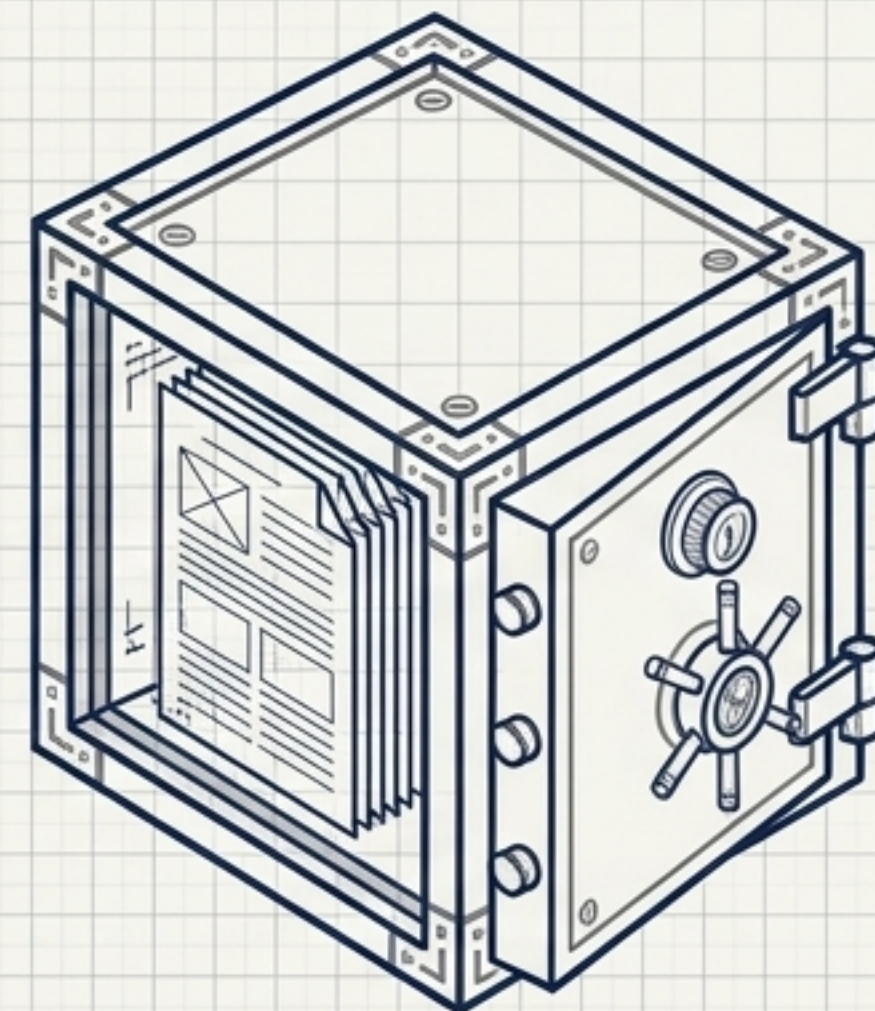
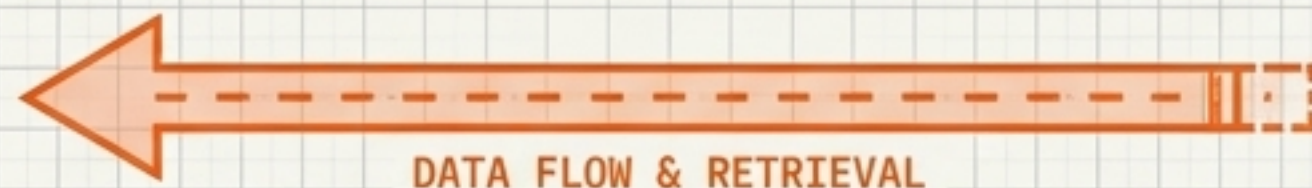
Answers only from real Holberton documents. No hallucinations. Instant retrieval.

# LANGUAGE MODELS ARE EXCELLENT AT READING BUT TERRIBLE AT REMEMBERING PRIVATE DATA



## LARGE LANGUAGE MODEL

Excellent at summarizing.  
Lacks your private data.



## RETRIEVAL-AUGMENTED GENERATION (RAG)

Gives the model a trusted,  
searchable memory.

## INDUSTRY DEPLOYMENTS



**CUSTOMER SUPPORT:**  
Answering from  
product docs



**LEGAL TECH:**  
Searching case law  
before drafting



**HR SYSTEMS:**  
Parsing company  
handbooks



**MEDICAL AI:**  
Retrieving clinical  
guidelines

# THE RAG ARCHITECTURE REQUIRES TWO DISTINCT PROCESSING PIPELINES

[LANE 1:  
OFFLINE INDEXING]

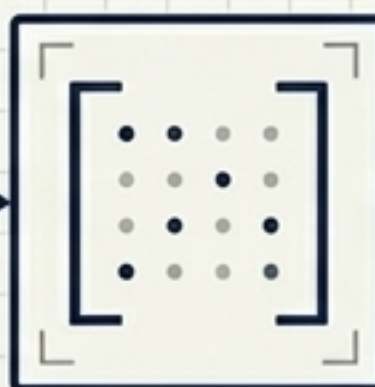
(Runs Once)



91 DOCUMENTS



CHUNKING



VECTOR  
CONVERSION



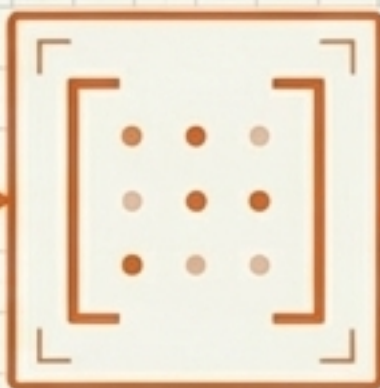
STORE IN  
VECTOR DB

[LANE 2:  
LIVE QUERY]

(Runs on Every Question)



USER  
QUESTION



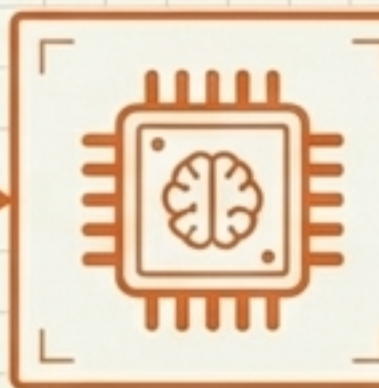
CONVERT TO  
VECTOR



SEARCH  
CLOSEST MATCH



RETRIEVE  
CHUNKS



GENERATE  
ANSWER

**QUESTION → EMBED → SEARCH → RETRIEVE → GENERATE → ANSWER**

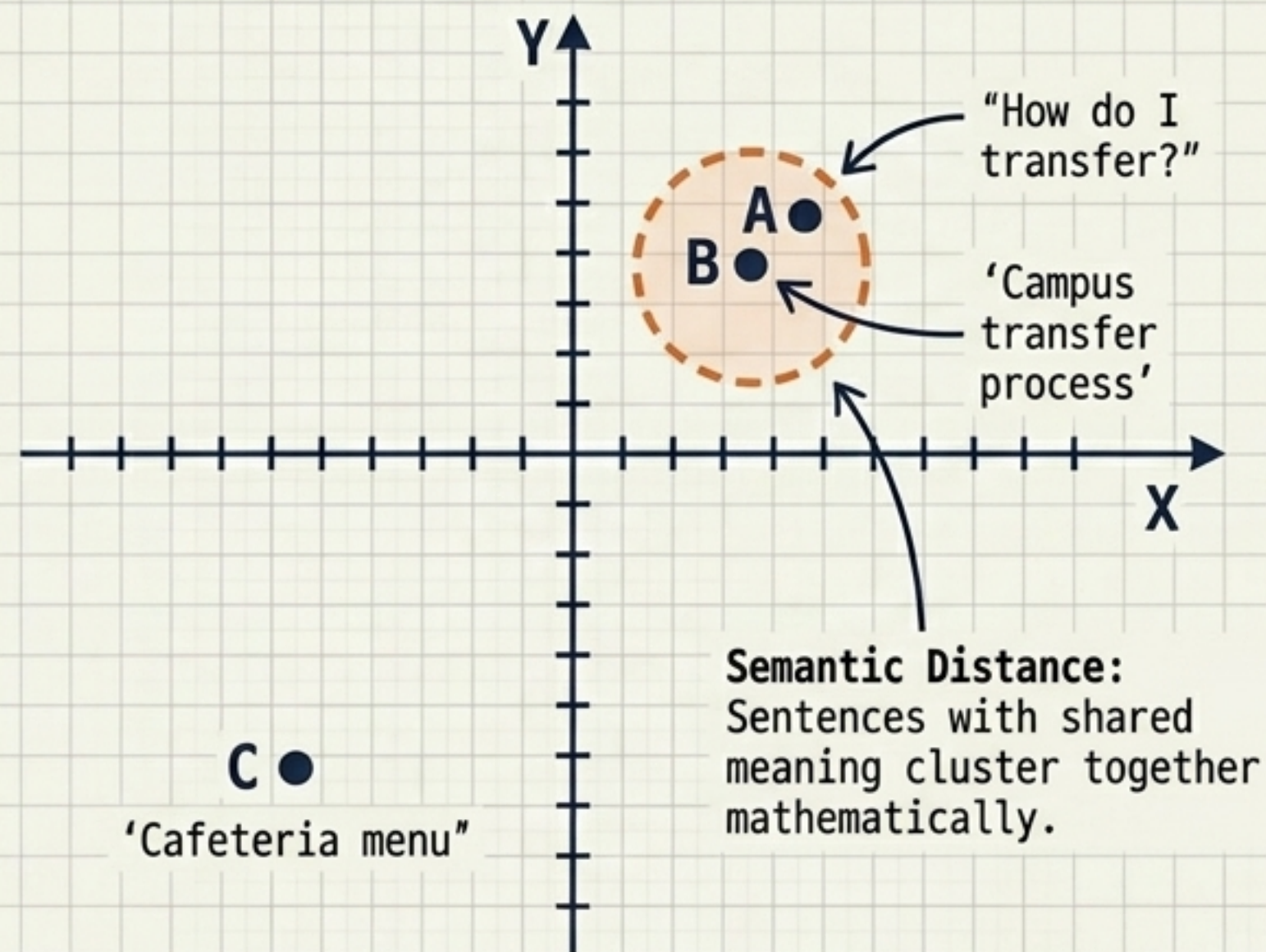
# VECTOR EMBEDDINGS MAP HUMAN LANGUAGE LANGUAGE INTO MATHEMATICAL PROXIMITY

## THE SEARCH MATRIX

|                 |          |   |             |
|-----------------|----------|---|-------------|
| KEYWORD SEARCH  | Transfer | ✗ | Eligibility |
| SEMANTIC SEARCH | Transfer | ✓ | Eligibility |

Matches underlying meaning,  
bypassing vocabulary constraints.

## THE EMBEDDING SPACE MAP



Model: Google gemini-embedding-001 | Status: Free, state-of-the-art

# FAISS EXECUTES SEMANTIC SIMILARITY SEARCHES IN MILLISECONDS WITHOUT CLOUD INFRASTRUCTURE

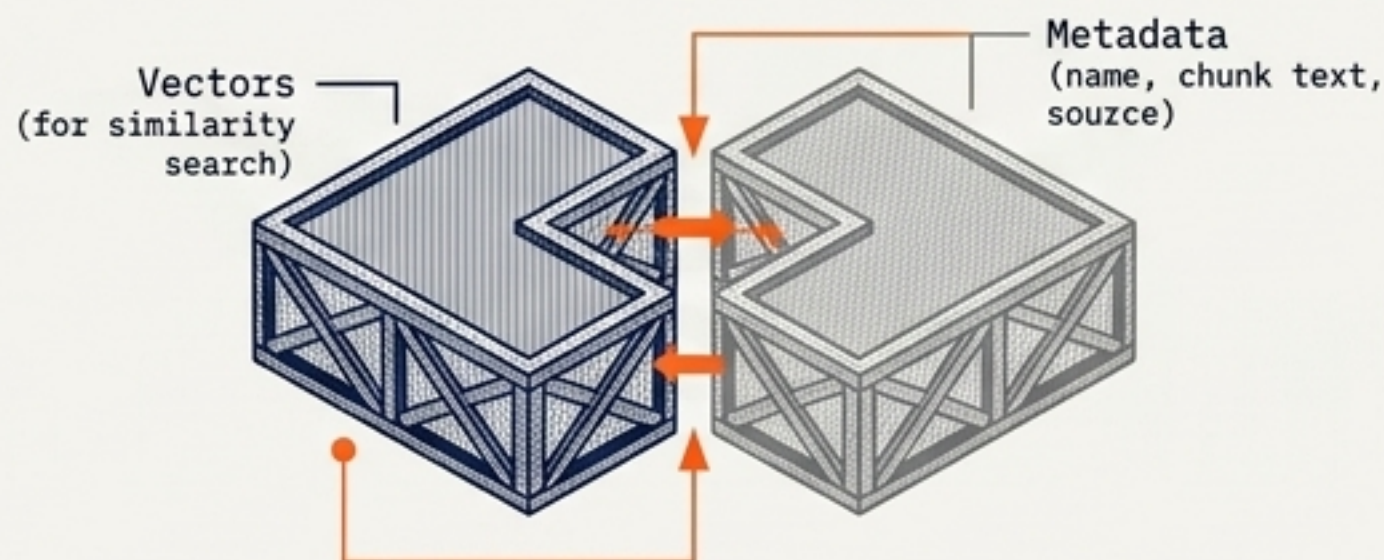
Component Spec: FAISS  
(Facebook AI Similarity Search)

- Open source
- Runs completely locally
- Zero cloud compute cost
- Production-tested by Meta & Spotify

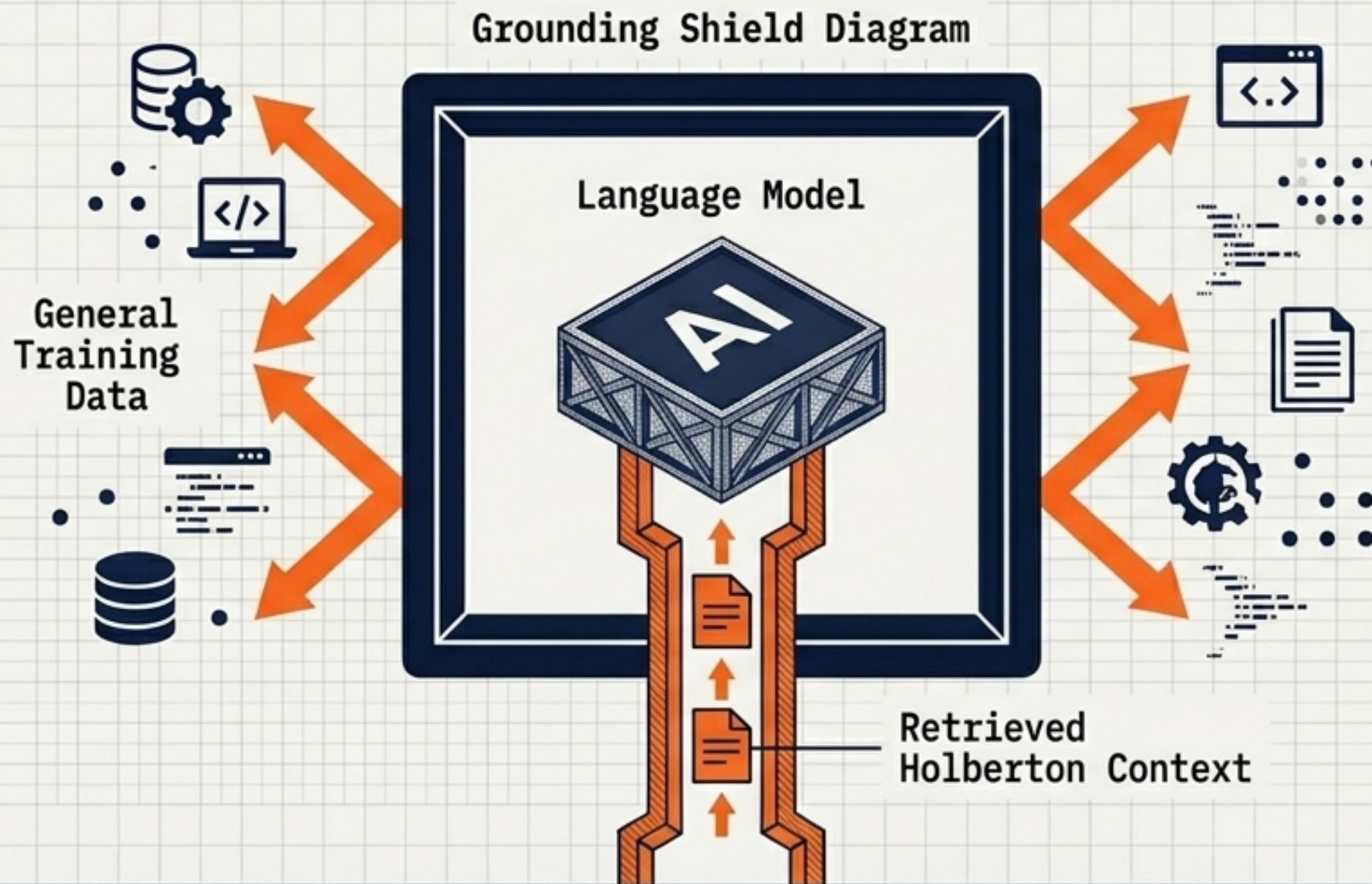
Scanning 91 Holberton Documents

100%

## Query Speed: 12ms



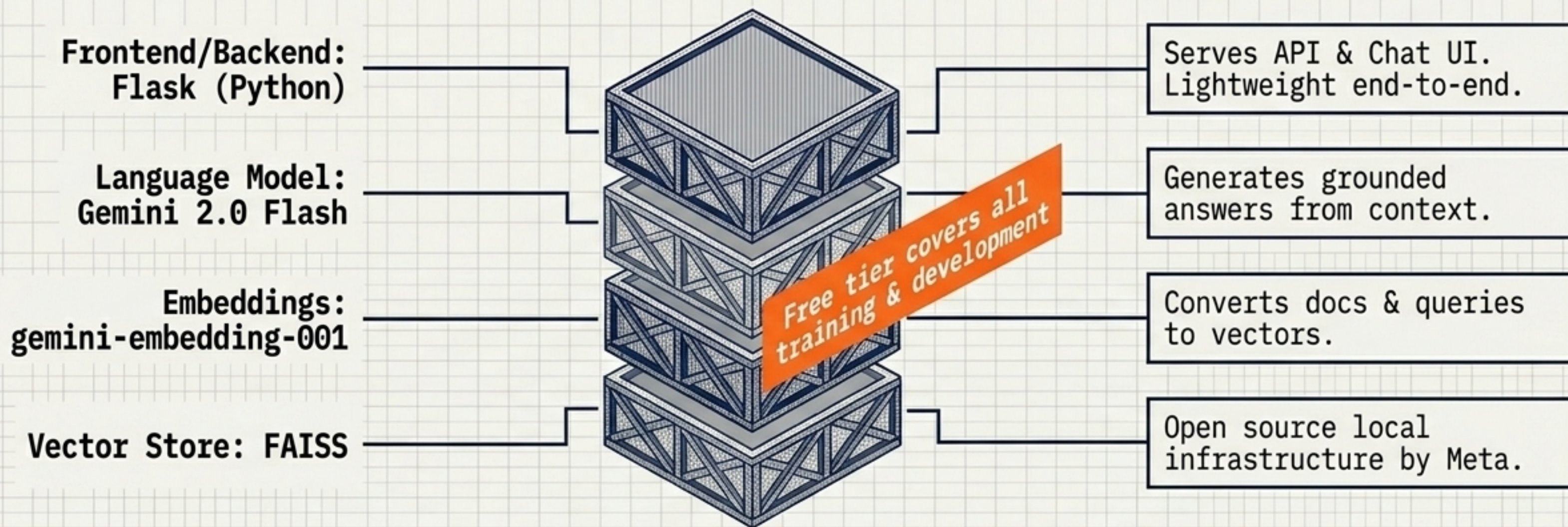
# GROUNDING RESTRICTS THE LANGUAGE MODEL TO ONLY USE RETRIEVED CONTEXT



The model reads the retrieved chunks and writes a grounded answer. It is strictly instructed to ignore external training data.

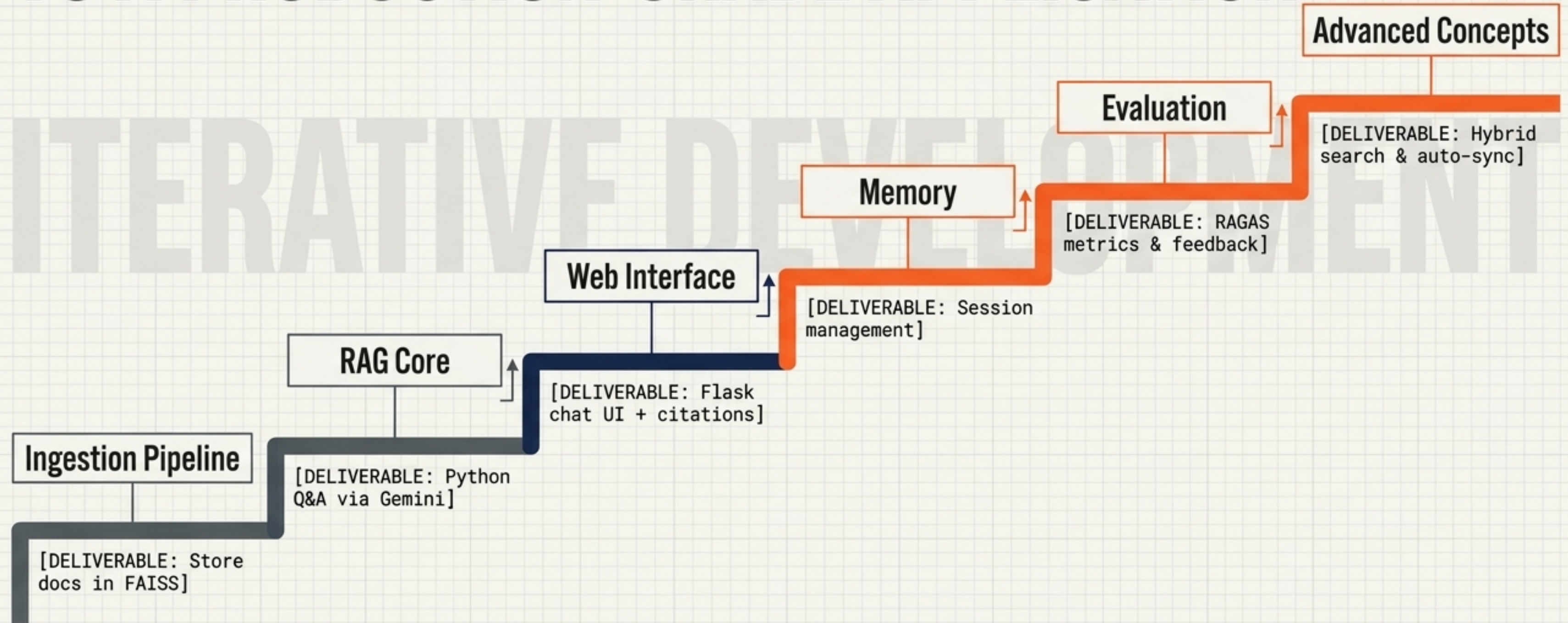
|         |                                                    |
|---------|----------------------------------------------------|
| Input:  | "What happens if I miss a project deadline?"       |
| Action: | Model processes ONLY the retrieved syllabus chunk. |
| Output: | Cited answer preventing hallucination.             |

# AN ENTERPRISE-GRADE ARCHITECTURE BUILT ENTIRELY ON A FREE AND LOCAL TECHNOLOGY STACK

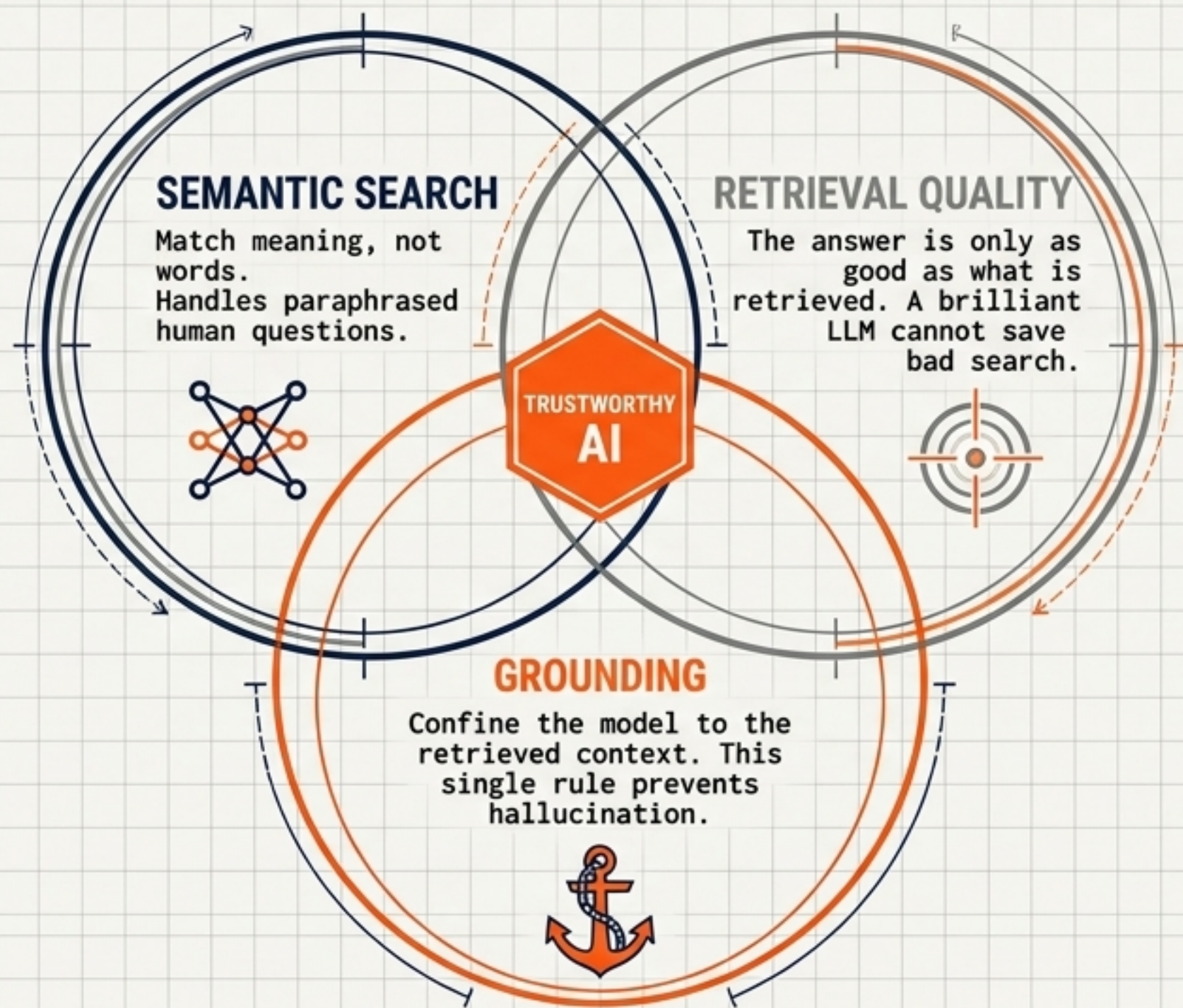


Every component mirrors industry standards at scale, running entirely on a single laptop.

# EVOLVING THE SYSTEM FROM A BASIC SCRIPT TO A PRODUCTION-GRADE APPLICATION



# THE RAG TRINITY: THREE PRINCIPLES THAT DICTATE SYSTEM SUCCESS



# YOU JUST BUILT THE EXACT ARCHITECTURE FORTUNE 500 COMPANIES PAY MILLIONS TO DEPLOY

| ARCHITECTURE LAYER | HOLBERTON LOCAL STACK | AWS                     | GOOGLE CLOUD     | MICROSOFT AZURE      |
|--------------------|-----------------------|-------------------------|------------------|----------------------|
| VECTOR SEARCH      | FAISS                 | BEDROCK KNOWLEDGE BASES | VERTEX AI SEARCH | AZURE AI SEARCH      |
| EMBEDDINGS         | GEMINI EMBEDDINGS     | TITAN EMBEDDINGS        | GECKO            | OPENAI ADA           |
| LANGUAGE MODEL     | GEMINI FLASH          | CLAUDE 3                | GEMINI PRO       | GPT-4O               |
| GENERATION RULE    | GROUNDING PROMPTING   | BEDROCK GUARDRAILS      | VERTEX GROUNDING | AZURE CONTENT SAFETY |

Understanding the primitives—embeddings, vector search, and evaluation—gives you the power to build and improve **any managed cloud platform**.

This is **not an academic exercise**. This is the exact pattern the AI industry is hiring for right now.